WIC during pregnancy and children's 1997 mathematics raw scores

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Introduction

This paper examines the relationship between prenatal participation in the Women, Infant, and Children's Nutrition Program and children's math achievement, as measured by the 1997 Woodcock–Johnson Revised Math raw score (*mathraw97*). The analysis also considers whether prenatal WIC participation, as an indicator of early nutritional intervention, has lasting impacts on children's cognitive outcomes by accounting for controls in child age, *AGE97*; family income, *faminc97*; and low-birth-weight status, bthwht. By using data from the Child Development Supplement to the Panel Study of Income Dynamics (PSID-CDS), this analysis situates WIC participation within a broader socioeconomic and developmental context, in which both economic resources and health at birth may combine to influence educational trajectories.

Besides direct testing of the relationship between WIC and math achievement, this report also investigates model quality through a series of assumption diagnostics and correction procedures. More specifically, we check whether the general linear model assumptions of linearity, homoscedasticity, and normality of residuals hold for the specified regression model and investigate potential outliers or influential cases distorting the inference. Finally, we examine whether the addition of parenting practices at home (*HOME97*) increases the explanatory power of the model. By comparing results before and after model respecification, this report provides both statistical rigor and policy insight into how family and developmental factors jointly contribute to children's cognitive outcomes.

Methods

Data Source

This analysis uses data from the Child Development Supplement (CDS) to the Panel Study of Income Dynamics (PSID), which is a nationally representative longitudinal dataset that

links family socioeconomic characteristics to children's developmental and academic outcomes. The sample includes the 3- to 13-year-old children in 1997 with complete information on math achievement, WIC participation, birth weight, family income, and parenting environment.

Variables

- **Dependent variable:** mathraw97, the child's raw score on the Woodcock-Johnson Revised Mathematics Achievement Test, ranging from 0 to 98.
- **Key independent variable:** WICpreg indicates whether the child's mother participated in the Women, Infant, and Children (WIC) nutrition program during pregnancy (1 = Yes, 0 = No).
- **Control variables:**
  - *AGE97* (years): child's age in 1997 (3–13)
  - *faminc97*: family income for 1997, in constant 2002 U.S. dollars (-$72,296 to $784,610).
  - *bthwht*: low birth weight indicator (1 = low birth weight, 0 = non-low birth weight).
  - *HOME97*: This is a composite measure of emotional and cognitive stimulation at home, ranging from 7 to 27, and it is used later in this paper to test incremental validity.

Analytical Strategy

Analysis follows a two-stage modeling approach:

1. **Baseline model:**

$$mathraw97 = \beta_0 + \beta_1 AGE97 + \beta_2 faminc97 + \beta_3 bthwht + \beta_4 WICpreg + \varepsilon$$

This model estimates the association between prenatal WIC participation and child math achievement, controlling for age, income, and birth weight.

2. **Revised model:**

$$mathraw97 = \beta_0 + \beta_1 AGE97_C + \beta_2 \,\text{asinh}(faminc97_C) + \beta_3 bthwht + \beta_4 WICpreg + \beta_5 HOME97_C + \varepsilon$$

This revised specification incorporates a variance-stabilizing transformation for family income, inverse hyperbolic sine; centers the continuous variables; and adds *HOME97* to examine whether parenting practices mediate or account for additional variation in achievement.

Diagnostic Tests and Corrections

Model assumptions were checked by the following diagnostics:

- **Linearity**: Component-plus-residual plots (partial residual plots) check whether relationships between continuous predictors and the outcome are linear.
- **Homoscedasticity**: Breusch-Pagan and White tests check whether residual variance is constant across predicted values.
- **Normality**: Histograms and Q-Q plots of residuals check for normal distribution of errors.
- **Influence and outliers:** Cook's distance, leverage, and DFBETAs identify observations with disproportionate influence on parameter estimates.

Where violations of assumptions were detected, corrective actions included transformation of faminc97 to reduce skewness, addition of HOME97 to capture omitted variance, mean-centering of continuous variables for interpretability, and use of heteroskedasticity-robust standard errors, HC3, for valid inference under non-constant variance.

Results

Descriptive Statistics

Descriptive analysis, shown in Table 1, provides an overview of the data's central tendencies, dispersion, and relationships among variables. The mean mathematics raw score (*mathraw97*) was roughly 36.3 (SD = 22.3), with values ranging from 0 to 98. The average age of children (AGE97) was 7.47 years old (SD = 2.93, range 3-13). Family income in 1997 (*faminc97*) had a mean of $49,800 (SD = $49,800), but with a wide range from -$72,296 to $784,610, indicating strong right-skewness. The *HOME97* parenting composite averaged 18.9 (SD = 3.6, range 7-27). Among binary variables, 40.4% of the mothers reported participating in WIC during pregnancy (*WICpreg*), and 38.8% of children were classified as low birth weight (*bthwht* = 1).

Table 1. *Descriptive Statistics*

STATISTIS	MATHRAW97	AGE97	FAMINC97	HOME97
VARS	1.0000000	2.00000000	3.000000	4.0000000
N	2211.0000000	2223.00000000	3563.000000	3563.0000000
MEAN	36.3265491	7.46693657	49841.254516	18.9230143
SD	22.2712427	2.93181744	49751.071933	3.6229133
MEDIAN	37.0000000	7.00000000	39118.440000	19.2000000
TRIMMED	35.8111928	7.45025295	42475.845780	19.0170116
MAD	29.6520000	4.44780000	31740.479316	4.1512800
MIN	0.0000000	3.00000000	-72296.260000	7.0000000
MAX	98.0000000	13.00000000	784610.590000	27.0000000
RANGE	98.0000000	10.00000000	856906.850000	20.0000000
SKEW	0.1230937	0.04350468	4.674435	-0.2591128
KURTOSIS	-1.1241880	-1.24520501	43.307937	-0.5845412
SE	0.4736419	0.06218236	833.478749	0.0606946

The pairwise correlations, displayed in Table 2, show *mathraw97* is highly correlated with *AGE97* ($r \approx 0.92$) and then has a moderate correlation with *HOME97* ($r \approx 0.31$) and *faminc97* ($r \approx 0.15$). This thus reflects that math performance should be greater with increased age, enriched home environment, and increased family incomes.

Table 2. *Paired Correlations*

	mathraw97	AGE97	faminc97	HOME97
MATHRAW97	1.0000000	0.9170191	0.1532505	0.3119211
AGE97	0.9170191	1.0000000	0.0501276	0.1974595
FAMINC97	0.1532505	0.0501276	1.0000000	0.3031655
HOME97	0.3119211	0.1974595	0.3031655	1.0000000

Baseline Model (Before Diagnostics)

This OLS regression estimated the baseline association between child math scores and the four predictors, including age, family income, birth weight, and WIC participation. The model explains approximately 86.5% of the variation in math scores, using $R^2 \approx 0.865$. Results indicated that *AGE97* was a strong and statistically significant predictor, as evidenced by $\beta \approx 7.01$, $p < .001$, indicating that with every additional year of age, math raw scores increased by about seven points. Family income was positively related to achievement, $\beta \approx 0.000032$, $p < .001$, meaning that a child's math score increased by approximately 0.32 points for every $10,000 increase in family income. Low birth weight negatively affected math performance, $\beta \approx -2.15$, $p < .001$; similarly, WIC participation during pregnancy also showed a small negative relationship, $\beta \approx -3.11$, $p < .001$. These baseline findings suggest that socioeconomic and early health factors significantly affect academic outcomes.

Table 3. *Baseline Model*

Regression Results

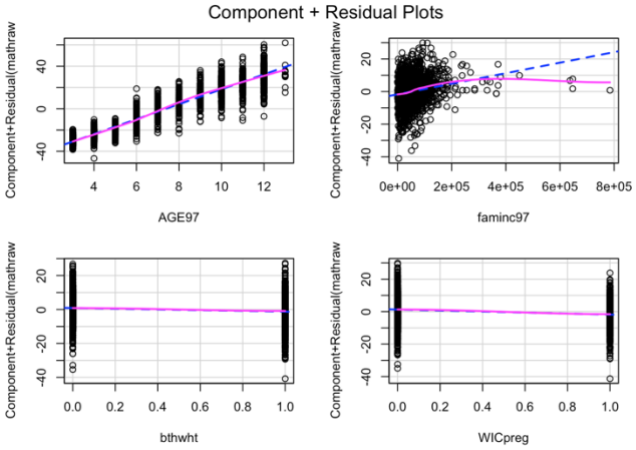
	Dependent variable:
	mathraw97
AGE97	7.013***
	(0.064)
faminc97	0.00003***
	(0.00000)
bthwht	-2.149***
	(0.383)
WICpreg	-3.113***
	(0.402)
Constant	-15.548***
	(0.586)
Observations	2,042
R^2	0.865
Adjusted R^2	0.865
Residual Std. Error	8.204 (df = 2037)
F Statistic	3,257.124*** (df = 4; 2037)
Note:	*p<0.1; **p<0.05; ***p<0.01

Diagnostic Evaluation

To assess whether assumptions of the general linear model were met, four major diagnostic checks were done: linearity, homoscedasticity, normality of residuals, and influence/outliers.

Linearity: Partial residual plots were used to explore whether each continuous predictor had a linear relationship with *mathraw97*. For *AGE97*, the plot confirmed a clear, positive linear trend; this is expected, as older children will always do better on raw math scores. For *faminc97*, the plot exhibited curvature and non-constant variance: middle-income children seemed to benefit most from increased family income, but very high-income values showed diminishing returns. This was an indication of a nonlinear relationship, which argued for a transformation such as asinh or log that produced a more stable fit.

Figure 1. *Linearity (component + residual plots)*



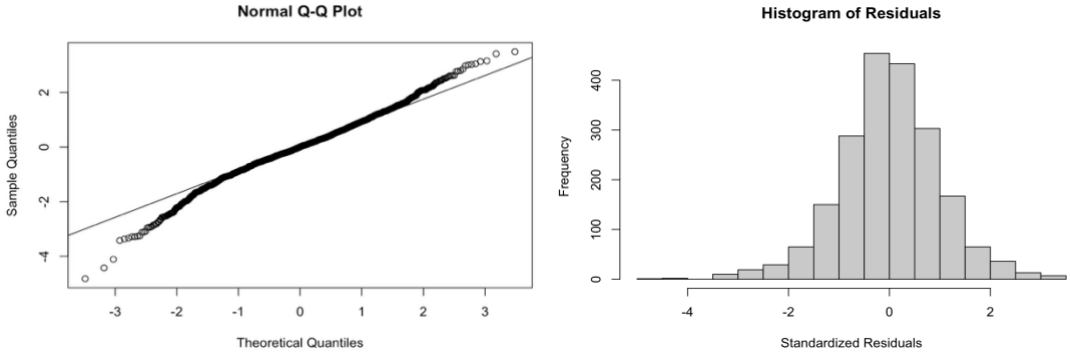
Homoscedasticity: The Breusch-Pagan and White tests were conducted to check for constant variance of residuals. Both tests rejected the null hypothesis of homoscedasticity, $p < .001$, indicating that residual variance increased with predicted math scores. In fact, this heteroskedasticity suggests that there might be downward bias in the OLS standard errors, hence motivating the use of HC3 robust standard errors to correct the inference.

Table 4. *Breusch–Pagan and White tests*

Test <chr>	BP <dbl>	df <dbl>	p_value <dbl>
Breusch-Pagan	164.9294	4	1.280940e-34
White Test	157.7684	2	5.508342e-35

Normality: Residual normality has been checked using histograms and Q-Q plots. The histogram of standardized residuals was approximately but not perfectly bell-shaped, with mild right skewness. The Q-Q plot showed heavier tails than for a normal distribution, with several points falling far off at either extreme. Given the large sample size, these deviations from perfect normality do little to threaten the validity of estimates obtained from the model. Having said that, normality improved subsequent to the application of the asinh transformation to income and the inclusion of *HOME97*.

Figure 2. *Normality of residuals*



Influence and Outliers: Potential influential cases were examined by Cook's Distance, leverage values, and DFBETAs. Overall, a few observations had leverages greater than the conventional threshold (2k/n), and several had moderately high Cook's D values of approximately 4/n. However, no case had an undue influence on overall estimates. A sensitivity analysis excluding high-leverage points resulted in coefficient magnitudes and directions that are almost identical, indicating that the model results are insensitive to outliers.

Table 5. *Influence Measures*

dfb.bthw <dbl>	dfb.WICp <dbl>	dffit <dbl>	cov.r <dbl>	cook.d <dbl>	hat <dbl>
1.714664e-03	9.306282e-04	3.037972e-03	1.0051023	1.846758e-06	0.002639175
-2.195779e-02	2.959290e-02	-6.032285e-02	1.0029775	7.277280e-04	0.003251352

1.901606e-02	3.755392e-02	6.427371e-02	1.0032450	8.261627e-04	0.003591760
-1.963171e-02	7.893865e-03	-5.137251e-02	1.0039774	5.278858e-04	0.003405049
-3.523452e-02	-2.689641e-02	-5.933098e-02	0.9999048	7.037483e-04	0.001925862
-7.351558e-03	-4.601962e-03	-1.271719e-02	1.0049186	3.236029e-05	0.002600345

Revised Model (After Corrections)

To address violations, the model was respecified to include the following: (1) the inverse hyperbolic sine transformation of faminc97 to reduce skewness, (2) the addition of HOME97 to capture aspects of parenting practices, (3) mean-centering of continuous predictors, and (4) heteroskedasticity-robust (HC3) standard errors for inference. The revised model accounts for slightly more variation in math achievement, $R^2 \approx 0.869$, with stable coefficients across predictors. As before, *AGE97* remains the strongest determinant, $\beta \approx 6.89$, $p < .001$. asinh(faminc97) is positively associated with math scores, $\beta \approx 0.58$, $p < .01$, reflecting the benefits of higher income. *bthwht* continues to demonstrate a negative effect, $\beta \approx -1.93$, $p < .001$, while *WICpreg* maintains a negative but attenuated association, $\beta \approx -2.24$, $p < .001$. Notably, *HOME97* is now a significant positive predictor, $\beta \approx 0.69$, $p < .001$, in support of the hypothesis that enriched home environments enhance cognitive development.

Table 6. *Revised Model*

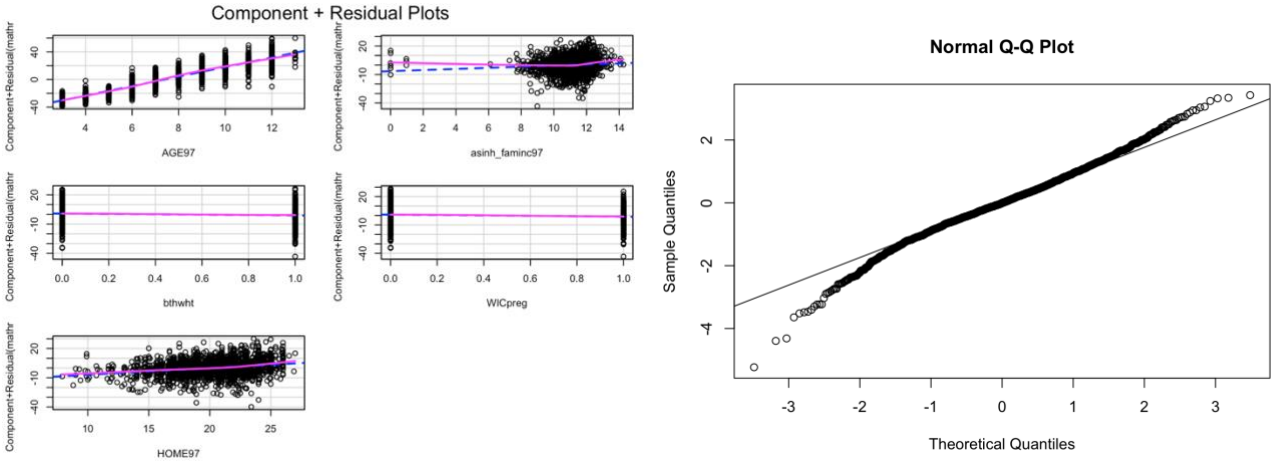
Regression Results

	Dependent variable:
	mathraw97
AGE97	6.893***
	(0.064)
asinh_faminc97	0.584***
	(0.171)
bthwht	-1.934***
	(0.377)
WICpreg	-2.237***
	(0.414)
HOME97	0.688***
	(0.069)
Constant	-33.805***
	(2.065)

Observations	2,042
R^2	0.869
Adjusted R^2	0.869
Residual Std. Error	8.066 (df = 2036)
F Statistic	2,709.592*** (df = 5; 2036)
Note:	*p<0.1; **p<0.05; ***p<0.01

Post-Correction Diagnostics After adjustments, model diagnostics improved significantly. Income reached linearity, and residuals had a more constant spread for most fitted values. Heteroskedasticity tests did remain significant, but this is to be expected for such large samples; robust SEs corrected the inference bias. The Q-Q plot showed less tail deviation, and influential points had minimal effects on the parameter estimates. Together, these corrections enhance both the statistical validity of the model and its interpretability, providing results more in line with theoretical expectations with respect to influences from socioeconomic and developmental statuses on children's math achievement.

Figure 3&4. *Component + Residual Plots and Q-Q Plots After Diagnostics*



Conclusion

This study investigated the relationship between prenatal participation in the Women, Infant, and Children Nutrition Program and children's mathematics achievement in 1997 using data from the PSID Child Development Supplement. The analysis verified that age, family income, and home environment are strong predictors of children's mathematics performance,

whereas low birth weight and WIC participation are linked to modestly lower scores. Diagnostic checks revealed that the homoscedasticity was violated at a moderate level and income did not have linear effects; these concerns were remedied through transforming income, using robust standard errors, and adding in the variable HOME97 to account for parenting behaviors. These modifications modestly improved model fit and clarified structure in relationships among socioeconomic, health, and environmental factors.

Overall, the results indicate that WIC participation is negatively related to math raw scores, but this most likely reflects selection effects-that is, families in which children participate in WIC face greater socioeconomic disadvantage rather than the program causing lower performance. The inclusion of HOME97 reduced the negative coefficient for WIC, suggesting that part of the observed association results from differences in the quality of the home learning environment. These findings reinforce the importance of considering family context when evaluating the effectiveness of policy interventions designed to enhance child outcomes. From an analytic standpoint, addressing assumption violations enhanced the reliability and interpretability of results and thus illustrates the payoff to careful diagnostics and respecification in applied regression modeling.

Appendix

```
good <- read.csv('/Users/leeeileen/Downloads/good.csv')
vars_needed <- c("mathraw97","AGE97","faminc97","bthwht","WICpreg","HOME97")

df_core <- good %>%
  select(all_of(vars_needed)) %>%
  mutate(across(everything(), as.numeric))
# Numeric descriptives
desc <- describe(df_core %>% select(mathraw97, AGE97, faminc97, HOME97))
# Binary counts and proportions
table(df_core$WICpreg, useNA = "ifany")
prop.table(table(df_core$WICpreg, useNA = "ifany"))
table(df_core$bthwht, useNA = "ifany")
prop.table(table(df_core$bthwht, useNA = "ifany"))
# Correlations
pair_cor <- cor(df_core %>% select(mathraw97, AGE97, faminc97, HOME97),
    use = "pairwise.complete.obs")
desc_t <- desc %>%
```

```
  mutate(Variable = rownames(desc)) %>%
  select(Variable, everything()) %>%
  pivot_longer(
    cols = -Variable,
    names_to = "Statistic",
    values_to = "Value"
  ) %>%
  pivot_wider(names_from = Variable, values_from = Value)
print(desc_t)
df_base <- df_core %>% drop_na(mathraw97, AGE97, faminc97, bthwht, WICpreg)
m_base <- lm(mathraw97 ~ AGE97 + faminc97 + bthwht + WICpreg, data = df_base)
summary(m_base)
# Heteroskedasticity (Breusch-Pagan and White tests)
bp1 <- bptest(m_base)
bp2 <- bptest(m_base, ~ fitted(m_base) + I(fitted(m_base)^2))
bp_results <- data.frame(
  Test = c("Breusch-Pagan", "White Test"),
  BP = c(bp1$statistic, bp2$statistic),
  df = c(bp1$parameter, bp2$parameter),
  p_value = c(bp1$p.value, bp2$p.value)
)
print(bp_results)
# Normality of residuals
qqnorm(rstandard(m_base)); qqline(rstandard(m_base))
hist(rstandard(m_base), main="Histogram of Residuals", xlab="Standardized Residuals")
# Outliers and influence
influence.measures(m_base)
plot(m_base, which = 5)
n <- nobs(m_base)
which(cooks.distance(m_base) > 4/n)
df_rev <- df_core %>%
  drop_na(mathraw97, AGE97, faminc97, bthwht, WICpreg, HOME97) %>%
  mutate(asinh_faminc97 = asinh(faminc97))
m_rev <- lm(mathraw97 ~ AGE97 + asinh_faminc97 + bthwht + WICpreg + HOME97, data =
df_rev)
summary(m_rev)
# Robust (HC3) standard errors
cov_hc3 <- vcovHC(m_rev, type = "HC3")
coeftest(m_rev, vcov. = cov_hc3)
# Diagnostics again
crPlots(m_rev)
bp_r_1 <- bptest(m_rev)
bp_r_2 <- bptest(m_rev, ~ fitted(m_rev) + I(fitted(m_rev)^2))
bp_results_2 <- data.frame(
  Test = c("Breusch-Pagan", "White Test"),
  BP = c(bp_r_1$statistic, bp_r_2$statistic),
  df = c(bp_r_1$parameter, bp_r_2$parameter),
  p_value = c(bp_r_1$p.value, bp_r_2$p.value)
)
print(bp_results_2)
qqnorm(rstandard(m_rev)); qqline(rstandard(m_rev))
summary(m_base)$r.squared
summary(m_rev)$r.squared
summary(m_base)$adj.r.squared
summary(m_rev)$adj.r.squared
anova(m_base, m_rev)
```